MILITARY SPECIFICATION

FILTER ELEMENTS, HYDRAULIC, DISPOSABLE, GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disposable hydraulic filter elements for use in ship hydraulic systems with an operating pressure up to 4000 pounds per square inch (lb/in^2) (276 BARs).

1.2 Classification. Filter elements shall be of the sizes specified in the specification sheet (see 3.1). The standard part numbers for the filter elements shall be as specified in the specification sheet (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

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SPECIFICATIONS

FEDERAL

PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

-MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-F-24702/1 - Filter Elements, Hydraulic, Disposable (Metric)
MIL-H-5606 - Hydraulic Fluid, Petroleum Base; Aircraft, Missle, and Ordnance.
MIL-H-17672 - Hydraulic Fluid, Petroleum, Inhibited.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-H-19457 - Hydraulic Fluid, Fire-Resistant, Non-Neurotoxic
MIL-H-22072 - Hydraulic Fluid, Catapult, Nato Code Number H-579.
MIL-R-83248 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.
MIL-R-83248/1 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant. O-Rings, Class 1, 75 Hardness.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging Requirements

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging, (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B93.22 - Method for Determining the Fabrication Integrity of a Hydraulic Fluid Power Filter Element.

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B93.24 - Method for Verifying the Flow Fatigue Characteristics of a Hydraulic Fluid Power Filter Element.
B93.25 - Hydraulic Fluid Power - Filter Elements - Verification of Collapse/Burst Resistance.
B93.31 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.
Y14.5 - Dimensioning and Tolerancing. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA, INC.
NAS 1638 - Cleanliness Requirements of Parts Used in Hydraulic System.

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, NW, Washington, DC 20036.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheet. The individual item requirements shall be as specified herein and in accordance with the specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Filter elements furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.3 Materials. Unless otherwise specified in the specification sheet, the filter elements and seals shall be constructed of materials that are compatible with fluids in accordance with MIL-H-19457, MIL-H-22072; and MIL-H-17672 (see 4.7.7).

3.3.1 Media materials. Filter element media materials shall be compatible with fluid containing up to 5 percent water. Such contaminated fluid shall not cause degradation of element performance.

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3.3.2 Imperfections. If the filter element is of the type wherein resinous or other material is used for elimination of imperfections, materials used shall be compatible with the hydraulic fluid, and not q ore than 2 percent of the filtering area shall be covered by the applied material.

3.3.3 Seals. Seals shall be synthetic rubber in accordance with MIL-R-83248 and MIL-R-83248/1.

3.3.4 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term “recovered materials” means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Construction.

3.4.1 Filter element. The filter element shall be cylindrical in shape and shall consist of a filter medium that shall be supported to withstand the side and end loads imposed during testing as specified in 4.7. The filter element shall be of the disposable type, designed for flow from outside in Each element shall be equipped with seals as specified in MIL-R-83248.

3.4.2 Dimensions. The filter element dimensions shall be as specified in the specification sheet.

3.4.3 Element covering. The element shall be constructed with an outer wrap of metal screen or perforated metal to protect the filter media pack from damage during handling.

3.5 Performance.

3.5.1 Fabrication integrity. (bubble point). The filter element’s bubble point shall be equal to or greater than the value specified in the specification sheet when tested as specified in 4.7.1.

3.5.2 Filtration performance. Unless otherwise specified in the specification sheet, at least two filter elements shall be subjected to this test. The filter element when tested as specified in 4.7.2, shall exhibit the following performance values:

(a) Filtration ratio (beta ratio, B_x): a minimum beta ratio (see 6 4.1) equal to or greater than the value specified in the applicable specification sheet.
(b) Dirt holding capacity: normalized capacity (see 6.4.2) equal to or greater than the value specified in the applicable specification sheet at a terminal pressure drop (see 6.4.4) of 40 lb/in^2 (2.76 BARs).
(c) Clean element pressure drop: at rated flow of fluid in accordance with MIL-H-17672 a clean element pressure drop equal to or less than the value specified in the specification sheet

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3.5.3 Flow fatigue. The filter element shall withstand, without damage, the number of flow fatigue pressure cycles specified in the specification sheet (see 4.7.3).

3.5.4 Collapse pressure. The filter element shall withstand, without structural damage, the collapse pressure equal to or greater than the value specified in the specification sheet (see 4.7.4).

3.5.5 Reverse flow. The filter element shall withstand, without damage, reverse differential pressure surges and flow to 15 lb/in^2 (see 4.7.5),

3.5.6 Media migration. There shall be no media migration from the filter element (see 4.7.6).

3.6 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.7 Identification. Each filter element shall be clearly and permanently identified on the end cap and the following information shall be provided:

(a) Filter element part number (see 1.2).
(b) Manufacturer's part or drawing number.
(c) Manufacturer's name or trademark.
(d) Non-cleanable.

3.8 Workmanship. Where dimensions and tolerances may affect the interchangeability or performance of the filter, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a Part of the contractor’s overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4,3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to the Command or agency concerned. Qualification inspection shall consist of the examination and tests specified in table I. For each element, tests shall be performed in the sequence shown in table I. Failure of any test shall be cause for rejection.

TABLE I. Qualification inspection.

Examination and tests	Requirement paragraph	Test paragraph
Element nos. 1 and 2 (Least bubble point elements)		
Examination	3.3, 3.4.1, 3.6, 3 7, and 3.9	4.6
Bubble point	3.5.1	4 7.1
Filtration performance	3.5.2	4.7 2
Element collapse pressure	3.5.4	4 7 4
Bubble point	3.5.1	4.7.1
Element no. 3 (3rd lowest bubble point element)		
Examination	3.3, 3 4.1, 3.6, 3.7, and 3.9	4.6
Bubble point	3.5.1	4.7.1
Flow fatigue	3.5.3	4 7 3
Bubble point	3.5 1	4.7.1
Element no. 4 (4th lowest bubble point element)		
Examination	3.3, 3 4.1, 3 6, 3.7, and 3 9	4 6
Bubble point	3.5.1	4.7.1
Media migration	3.5.6	4 7 6
Reverse flow	3.5.5	4.7 5
Bubble point	3.5.1	4 7.1
Element nos. 5, 6 and 7 (Highest bubble point elements)		
Examination	3.3, 3.4.1, 3.6, 3 7, and 3.9	4 6
Bubble point	3.5.1	4.7 1
Compatibility	3.3.2	4.7 7
Bubble point	3.5.1	4 7 1
Additional tests	4 7 8	as required

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4.3.1 Drawings. Prior to authorization for qualification testing, two sets of dimensional drawings shall be furnished with each filter element part numbers. Assembly drawings shall show a cutaway section of details in their normal assembly positions and shall carry part numbers of details and subassemblies. The following data shall be furnished on, or together with, the assembly drawings:

(a) Outline dimensions of the complete assembly.
(b) Bill of material, listing specifications, grade, and condition, or other data needed to identify the materials proposed.
(c) General notes stating complete performance data including rated flow capacity, pressure drop data, pressure ratings, filtration rating, and dirt-holding capacity.
(d) Unless such information is provided in the qualification test report, filter element drawings shall include a cross sectional view with each individual component identified Proprietary information, such as number of square inches of media or composition of media, is not required.

Dimensioning and tolerancing on drawings shall be in accordance with ANSI Y14.5.

4.3.2 Retention of qualification. Retention of qualification for filter elements shall be dependent on a 5-year verification of continued compliance with the requirements of this specification. Filter performance test specified in 4.7.2 shall be performed to verify the continued compliance with the specification requirements.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 and 4.4.2.

4.4.1 Individual inspection. The individual inspection specified in table II shall be conducted on each element. Any filter element containing a defect shall be rejected.

TABLE II. Quality conformance inspection.

Individual inspection		
Examination and tests	Requirement paragraph	Test paragraph
Examination Bubble point	3.3, 3.4.1, 3.6, 3.7 and 3.8 3.5 1	4.6 4 7 1
Sampling		
Clean element pressure drop	3.5.2	4.7.2

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4.4.2 Sampling tests. A sample shall be selected from each inspection lot in accordance with MIL-STD-105 using special inspection level S-1. The sampling tests specified in table II shall be conducted on each sample filter element Failure of the test shall be cause for rejection of the filter element inspection lot.

4.5 Test conditions.

4.5.1 Test fluid. Hydraulic fluid conforming to MIL-H-5606 shall be used for the filtration performance test (see 4.7.2) and shall be with or without antistatic additives as specified in the specification sheet. Unless otherwise specified (see 6.2.1), other tests may be run with either MIL-H-5606 fluid or one of the fluids compatible with the filter media identified in 3.3. If a separate clean element pressure drop test is not run with MIL-H-17672 fluid, the calculated pressure drop shall be determined with MIL-H-5606 fluid. Unless otherwise approved by the qualifying activity, this calculation shall be based on the ratio of the viscosities of the two fluids.

4 5.1.1 Test fluid cleanliness. For quality conformance tests and critical qualification tests, the test fluid shall be precleaned to a cleanliness level equal to 4 in accordance with NAS 1638.

4.5.2 Test fluid temperature. Where fluid in accordance with MIL-H-17672 is used for testing, the fluid temperature shall be as specified in the applicable specification sheet, and the temperature shall be maintained to plus or minus 2 degrees Celsius (°C). Viscosity data used to determine test temperature shall be included in the test report. Unless otherwise specified in the specification sheet, where fluid in accordance with MIL-H-5606 is used for testing, the actual temperature of the test fluid shall be 40 + 2°C.

4.5.3 Element housing. The test housing shall be as recommended by the filter manufacturer except that it shall be modified to block off the bypass circuit. The housing shall not permit fluid entry tangential to and at the circumference of the housing. Unless otherwise specified in the specification sheet, the inside volume of the housing shall not be in excess of two times the envelope volume of the element.

4.6 Examination. The filter elements shall be examined to determine compliance with the requirements specified herein with respect to construction design, dimensions, visible defects, marking, and workmanship.

4.7 Test methods.

4.7.1 Bubble point (fabrication integrity). Filter elements shall be tested as specified in ANSI B93 22 to determine the initial bubble point. A bubble point pressure lower than that specified in 3 5.1, shall constitute failure of this test.

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4.7.1.1 Test point values. The initial bubble point value of the six qualification test elements of each size shall be determined prior to further testing. The minimum initial bubble point pressure for each production element shall be established based on the bubble points of the qualification test elements. The minimum bubble point established shall be 90 percent of the minimum bubble point of the qualification test element with the lowest bubble point, but shall be not less than 4.5 inches of water.

4.7.2 Filtration ratio, dirt holding capacity, and pressure drop. The filter elements shall be tested as specified in ANSI B93.31 to determine filtration ratio, dirt capacity, and pressure drop of the elements. The dirt capacity shall be determined at a terminal pressure drop (see 6 4.4) of 40 lb/in^2 (2.76 BARs).

4.7.3 Flow fatigue. Filter elements shall be installed in nonbypass type filter housings and shall be subjected to pressure-flow cycles as specified in ANSI B93.24. A cycle shall consist of increasing the differential pressure across the filter element from zero to the maximum specified and back to zero by first increasing, then decreasing, the flow through the test filter element which has been loaded with A-C fine or A-C coarse dust, or equivalent. The maximum cycle rate shall be 60 cycles per minute. There shall be no evidence of damage as a result of this test, and, after testing the element bubble point shall be not less than 80 percent of the initial bubble point of the element prior to testing.

4.7.4 Filter element collapse pressure. The filter elements shall be tested as specified in ANSI B93.25 to determine the collapse resistance pressure of the elements. Nonconformance to the requirements of 3.5.4 shall constitute failure of this test. The bubble point after testing shall be not less than 80 percent of the initial bubble point value of the element prior to testing.

4.7.5 Filter element reverse flow test. With rated flow in the reverse direction from the element collapse test (see 4,7,4), the element shall be loaded with A-C fine or A-C coarse test dust until a differential pressure of 15 lb/in^2 is obtained across the element. The element shall withstand the reverse flow differential pressure without structural damage. The initial bubble point of the element after reverse flow testing shall be not less than 80 percent of the initial bubble point of the element prior to reverse flow testing. Flow may be adjusted as necessary, to maintain the differential pressure of 15 lb/in^2 for at least 3 minutes.

4.7.6 Media migration and vibration analysis of filter elements.

4.7.6.1 Preparation for analysis. The test element shall be installed in a filter housing or a special test housing with blanking connections installed. The housing shall be filled with prefiltered hydraulic test fluid.

4.7.6.2 Vibration test. The filter housing head shall be secured to the testing machine so that the vibration input is centrally aligned along the longitudinal axis of the filter element. The equipment shall be vibrated from 4 hertz (Hz) (or lowest attainable frequency) to 50 Hz in discrete frequency

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intervals of 1 Hz, at the amplitudes shown in table III. At each frequency, the vibration shall be maintained for times shown in table III, An acceptable alternative will be a continuous sweep from 4 to 50 Hz, over a period of at least 1 hour, with the amplitudes shown in table III.

TABLE III. Vibratory displacement single amplitude.

Frequency range (Hz)	Table amplitude 1/ (inches - minimum)	Time (minutes)
4 to 15	0.05	2
16 to 25	.04	1
26 to 33	.03	1
34 to 40	.02	1
41 to 50	.01	1

1/ For Government conducted tests, the maximum amplitude shall not exceed the minimum required amplitude by more than 20 percent.

4.7.6.3 Media migration analysis. The port blanks shall be removed and the filter and housing shall be flushed with 3000 milliliters (mL) minimum of prefiltered fluid and the effluent collected. The effluent shall be filtered through a membrane filter disk, 47 millimeters (mm) diameter, absolute 0.8 micrometer, type AA Millipore, or equal, which has been previously washed with prefiltered solvent, The membrane from this analysis and the membrane retained from the degree of filtration test (see 4,7.2) shall be examined for evidence of filter media migration. There shall be no media migration identifiable as coming from the filter element medium.

4.7.7 Material compatibility. Elements 4, 5, and 6 shall be tested for compatibility with fluid conforming to MIL-H-19457, MIL-H-22072 and MIL-H-17672 One patch (four required; each approximately 2 by 4 inches) of the filtration media shall also be tested for compatibility with each of above fluids. An untested fourth patch shall be retained for microscopic comparison with the immersed patches on completion of the tests. Test patches shall be taken from a single representative specimen of filtration media (approximately 8 by 4 inches) Each test fluid shall be thoroughly mixed with 5 percent water prior to testing, and stirred approximately every 24 hours thereafter, weekends excepted. During the test, water and fluid may be added as necessary to make up evaporation losses. Elements and testpatches shall be completely immersed in appropriate test fluids as designated above, for 195 hours at 85 to 90°C After immersion and decreasing of the elements and patches, there shall be no visible evidence of degradation or corrosion. Element bubble points shall be not less than 85 percent nor more than 120 percent of the bubble point at the beginning of the test. Patches shall be compared to the untested specimen under a microscope and visible difference noted.

4.7.8 Additional tests. After the fluid compatibility tests, element numbers 4, 5, and 6 may be subjected to any of the tests listed for element numbers 1, 2, or 3 and shall satisfactorily pass those tests. Tests to be conducted and elements selected shall be at the discretion of NAVSEA or its representative.

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4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

5.1.1.1 Lumber and plyood. Unless otherwise specified (see 6.2.1), all lumber plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

Level A and B - Type II - weather resistant.
Category I - general use.
Level C - Type I - non-weather resistant.
Category I - general use.

5 1.1.2 Fiberboard. Unless otherwise specified, fiberboard used in the construction of class-domestic, non-weather resistant fiberboard, and cleated fiberboard boxes shall meet the flame spread index and specific optic density requirements as specified in PPP-F-320.

5.2 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. Filter elements shall be cleaned and dried by a process and procedure in accordance with MIL-P-116 which will ensure removal of all foreign material without damage to the element. Unless otherwise specified (see 6.2.1), the cleaning and drying process and procedure shall be at the option of the contractor.

5.2.1.2 Unit protection. Each filter element shall be unit protected within a transparent bag in accordance with MIL-P-116, method IC-1. Each method IC-1 unit bag shall be supplemented with a snug-fitting water resistant folding, setup or metal edged paperboard or fiberboard box at the contractor's option, in accordance with the unit container requirements, table T , appendix F of MIL-STD-2073-1. Box closure shall be in accordance with the box specification or appendix thereto. Interior cushioning shall be provided, as necessary, to prevent element damage and puncture or damage to the protective unit bag. Unless otherwise specified (see 6.2.1), box options such as type, variety, grade and style shall be at the option of the contractor.

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5.2.2 Level C. Preservation shall be as specified for level A except:

(a) The unit protection shall be by method IC-3, bag closure accomplished by twist-tie with plastic clips or plastic coated ties, and the leak and heat-seal tests are not required.
(b) The unit containers specified in 5.2.1.2 shall be of the domestic/fire retardant or non-weather resistant type, class or variety. Method I using pressure-sensitive tape shall apply for fiberboard box closure.

5.2.3 Commercial. Commercial packaging (cleaning, preservation, cushioning and unit pack) requirements shall be in accordance with ASTM D 3951.

5.3 Packing. Packing when required (see 6.2.1), shall be level A, B, C, or commercial as specified (see 6.2.1).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, of uniform size, and contain identical quantities of identical filter elements.

5.3.2 Levels A, B, and C containers. Filter elements preserved as specified (see 5.2) shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with MIL-STD-2073-1. Unless otherwise specified (see 6.2.1), container selection shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2.1), level A shipping containers containing filter elements preserved level C or commercial shall be provided with waterproof caseliners in accordance with MIL-STD-2073-1.

5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with nonmetallic or tape banding and domestic or fire retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Filter elements preserved as specified (see 5.2) shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment and storage.

5 4 Palletized unit loads. When specified (see 6.2.1), containers shall be palletized in accordance with MIL-STD-2073-1.

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5.5 Marking.

5.5.1 Levels A, B, C, and commercial. In addition to any special marking required (see 6.2.1, 3.7, and herein), interior (unit) packs, shipping containers and palletized unit loads shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

6. NOTES

6.1 Intended use. Filter elements covered by this specification are intended for use in ship petroleum base oil (MIL-H-17672), water glycol (MIL-H-22072), and phosphate ester (MIL-H-19457) fluid hydraulic systems. For application to systems using other than these fluids, users are cautioned to insure compatibility with filter element specified herein.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Filter element part number (see 1.2).
(c) Title, number, and date of the applicable specification sheet (see 3.1).
(d) Test fluids, if other than specified (see 4.5.1).
(e) When fire retardant requirements are not required (see 5.1.1.1).
(f) Levels of presentation and packing required (see 5.2 and 5.3).
(g) Cleaning and drying if other than contractor's option (see 5.2.1.1).
(h) Box and container options if other than contractor's (see 5.2 1 1. and 5.3.2).
(i) When caseliners are not required (see 5.3.2. 1.1).
(j) When palletization is required (see 5.4).
(k) Special marking required (see 5.5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List QPL-24702 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 and information pertaining to qualification of products may be obtained from that activity Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

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6.4 Definitions. The following definitions apply throughout this specification.

6.4.1 Filtration ratio (beta, β_x). Filtration ratio is the ratio of number of particles greater than a given size (x) in the fluid entering the filter element to the number of particles greater than the same size in the fluid leaving the filter element.

6.4.2 Normalized capacity. Normalized capacity is the apparent capacity divided by the rated flow gallons per minute of the element. The apparent capacity is the actual weight (grams) of the contaminant injected into the filter test system before the terminal pressure drop.

6 4.3 Rated flow. The rated flow is the maximum flow at which the element operates.

6.4.4 Terminal pressure drop. Terminal pressure drop is the maximum pressure drop permitted across the filter element before a change is required.

6 4.5 Pressure drop. Pressure drop is the difference in pressure across a filter element.

6.5 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Subject term (key word) listing.

Beta ratio
Bubble point
Collapse pressure
Element housing
Flow fatigue
Media migration

Preparing activity:
Navy - SH
(Project 4330-N046)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-F-24702(SH)	FILTER ELEMENTS, HYDRAULIC, DISPOSABLE, GENERAL SPECIFICATION FOR (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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